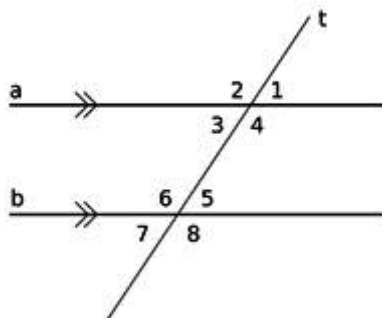


Lines and Angles

Parallel Lines and a Transversal

Parallel lines with transversal



Parallel lines with a transversal

- $\angle 1 = \angle 5$, $\angle 2 = \angle 6$, $\angle 4 = \angle 8$ and $\angle 3 = \angle 7$ (Corresponding angles)
- $\angle 3 = \angle 5$, $\angle 4 = \angle 6$ (Alternate interior angles)
- $\angle 1 = \angle 7$, $\angle 2 = \angle 8$ (Alternate exterior angles)

Lines parallel to the same line

- Lines that are parallel to the same line are also parallel to each other.

Introduction to Geometry

Angles and types of angles

When 2 rays originate from the same point at different directions, they form an angle.

- The rays are called arms and the common point is called vertex
- Types of angles : (i) Acute angle $0^\circ < a < 90^\circ$
 - (ii) Right angle $a = 90^\circ$
 - (iii) Obtuse angle : $90^\circ < a < 180^\circ$
 - (iv) Straight angle $= 180^\circ$
 - (v) Reflex Angle $180^\circ < a < 360^\circ$
 - (vi) Angles that add up to 90° are complementary angles
 - (vii) Angles that add up to 180° are called supplementary angles.

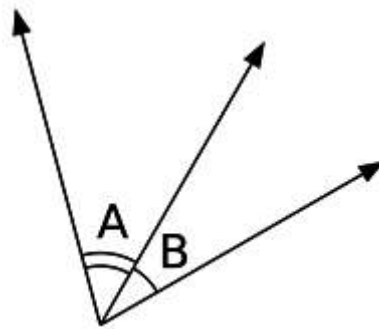
Intersecting Lines and Associated Angles

Intersecting and Non-Intersecting lines

- When 2 lines meet at a point they are called intersecting
- When 2 lines never meet at a point, they are called non-intersecting or parallel lines

Adjacent angles

2 angles are adjacent if they have the same vertex and one common point.



Adjacent angles

Linear Pair

When 2 adjacent angles are supplementary, i.e they form a straight line (add up to 180°), they are called a linear pair.

Vertically opposite angles

When two lines intersect at a point, they form equal angles that are vertically opposite to each other.

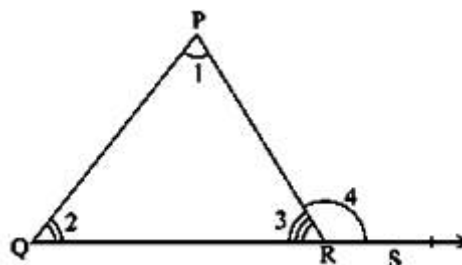
Basic Properties of a Triangle

Triangle and sum of its internal angles

Sum of all angles of a triangle add up to 180°

Exterior angle of a triangle = sum of opposite internal angles

- If a side of a triangle is produced, then the exterior angle so formed is equal to the sum of the two interior opposite angles



$\angle 4$ is the exterior angle

$$- \angle 4 = \angle 1 + \angle 2$$